

WHITE-COW FOMO MODEL

Independent V2 Evaluation Report

AI / IoT Based Smart Farm Perimeter Protection System

Evaluation Summary

Test Images	50
True Positive	28
True Negative	19
False Positive	1
False Negative	2
Accuracy	94.00%
Precision	96.55%
Recall	93.33%
F1 Score	94.92%
False Positive Rate	5.00%
Average Confidence	0.4775
Average Latency	2.061 ms

Confusion Matrix

	Predicted Cattle	Predicted Non-Cattle
Actual Cattle	28	2
Actual Non-Cattle	1	19

Model: one with white cow.eim

Input resolution: 160 × 160 RGB

Independent V2 set: 30 cattle + 20 non-cattle

Threshold: 0.50

MODEL DEVELOPMENT HISTORY

- Cleaned cattle dataset: 1,271 images.
 - Training split: 1,016 images.
 - Validation split: 255 images.
 - FOMO MobileNetV2 0.35.
 - 60 training cycles.
 - Learning rate: 0.001.
 - CPU training.
 - Data augmentation enabled.
 - INT8 quantized model.
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- Independent V2 evaluation set:
30 cattle images + 20 non-cattle images.
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- 350 greenery/background negative images
were added during model development.
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- Independent testing identified white/light
cattle as an important false-negative pattern.
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- White/light cattle examples were added.
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- Final white-cow model was evaluated again
on the same independent V2 test set.
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- Final accuracy: 94.00%
 - Final precision: 96.55%
 - Final recall: 93.33%
 - Final F1: 94.92%
 - Final FPR: 5.00%

CORRECT



Image: cattle_001.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 94.92%

Cattle Boxes: 2

Latency: 8.272 ms

Running Accuracy: 100.00%

CORRECT



Image: cattle_002.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 72.66%

Cattle Boxes: 2

Latency: 4.742 ms

Running Accuracy: 100.00%

CORRECT



Image: cattle_003.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 68.75%

Cattle Boxes: 1

Latency: 4.153 ms

Running Accuracy: 100.00%

WRONG



Image: cattle_004.jpg

Ground Truth: cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 3.768 ms

Running Accuracy: 75.00%

CORRECT



Image: cattle_005.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 96.48%

Cattle Boxes: 1

Latency: 2.610 ms

Running Accuracy: 80.00%

CORRECT



Image: cattle_006.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 96.48%

Cattle Boxes: 2

Latency: 2.090 ms

Running Accuracy: 83.33%

CORRECT



Image: cattle_007.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 92.97%

Cattle Boxes: 2

Latency: 2.114 ms

Running Accuracy: 85.71%

CORRECT



Image: cattle_008.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 89.84%

Cattle Boxes: 3

Latency: 1.871 ms

Running Accuracy: 87.50%

CORRECT



Image: cattle_009.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 92.97%

Cattle Boxes: 1

Latency: 1.905 ms

Running Accuracy: 88.89%

CORRECT



Image: cattle_010.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 89.84%

Cattle Boxes: 2

Latency: 1.907 ms

Running Accuracy: 90.00%

CORRECT



Image: cattle_011.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 64.45%

Cattle Boxes: 1

Latency: 1.642 ms

Running Accuracy: 90.91%

CORRECT



Image: cattle_012.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 64.45%

Cattle Boxes: 2

Latency: 1.863 ms

Running Accuracy: 91.67%

CORRECT



Image: cattle_013.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 50.00%

Cattle Boxes: 1

Latency: 2.124 ms

Running Accuracy: 92.31%

CORRECT



Image: cattle_014.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 91.41%

Cattle Boxes: 3

Latency: 1.957 ms

Running Accuracy: 92.86%

CORRECT



Image: cattle_015.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 76.56%

Cattle Boxes: 1

Latency: 1.704 ms

Running Accuracy: 93.33%

CORRECT



Image: cattle_016.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 76.56%

Cattle Boxes: 1

Latency: 1.708 ms

Running Accuracy: 93.75%

CORRECT



Image: cattle_017.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 82.81%

Cattle Boxes: 3

Latency: 1.671 ms

Running Accuracy: 94.12%

CORRECT



Image: cattle_018.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 76.56%

Cattle Boxes: 2

Latency: 1.608 ms

Running Accuracy: 94.44%

CORRECT



Image: cattle_019.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 87.89%

Cattle Boxes: 1

Latency: 1.706 ms

Running Accuracy: 94.74%

WRONG



Image: cattle_020.jpg

Ground Truth: cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.651 ms

Running Accuracy: 90.00%

CORRECT



Image: cattle_021.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 50.00%

Cattle Boxes: 3

Latency: 1.937 ms

Running Accuracy: 90.48%

CORRECT



Image: cattle_022.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 91.41%

Cattle Boxes: 1

Latency: 2.058 ms

Running Accuracy: 90.91%

CORRECT



Image: cattle_023.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 72.66%

Cattle Boxes: 1

Latency: 1.676 ms

Running Accuracy: 91.30%

CORRECT



Image: cattle_024.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 92.97%

Cattle Boxes: 2

Latency: 1.680 ms

Running Accuracy: 91.67%

CORRECT



Image: cattle_025.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 96.48%

Cattle Boxes: 3

Latency: 1.832 ms

Running Accuracy: 92.00%

CORRECT



Image: cattle_026.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 98.44%

Cattle Boxes: 4

Latency: 1.717 ms

Running Accuracy: 92.31%

CORRECT



Image: cattle_027.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 92.97%

Cattle Boxes: 6

Latency: 1.858 ms

Running Accuracy: 92.59%

CORRECT



Image: cattle_028.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 98.83%

Cattle Boxes: 1

Latency: 1.574 ms

Running Accuracy: 92.86%

CORRECT



Image: cattle_029.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 96.48%

Cattle Boxes: 2

Latency: 2.088 ms

Running Accuracy: 93.10%

CORRECT



Image: cattle_030.jpg

Ground Truth: cattle

Prediction: cattle

Confidence: 76.56%

Cattle Boxes: 1

Latency: 1.846 ms

Running Accuracy: 93.33%

CORRECT

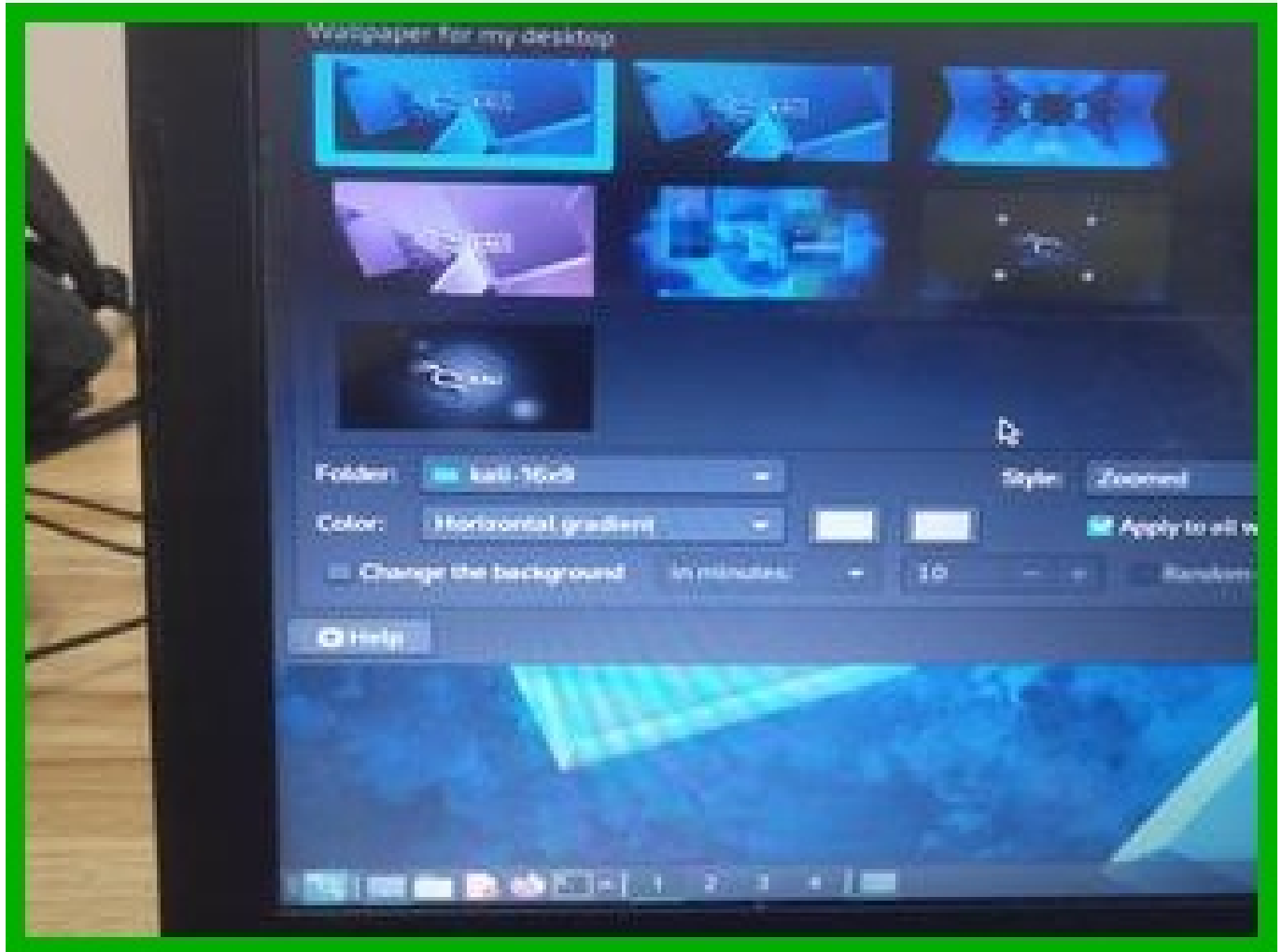


Image: noncattle_001.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.775 ms

Running Accuracy: 93.55%

CORRECT



Image: noncattle_002.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.517 ms

Running Accuracy: 93.75%

CORRECT



Image: noncattle_003.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.787 ms

Running Accuracy: 93.94%

CORRECT



Image: noncattle_004.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.822 ms

Running Accuracy: 94.12%

CORRECT



Image: noncattle_005.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.860 ms

Running Accuracy: 94.29%

CORRECT



Image: noncattle_006.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.495 ms

Running Accuracy: 94.44%

CORRECT



Image: noncattle_007.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.478 ms

Running Accuracy: 94.59%

CORRECT

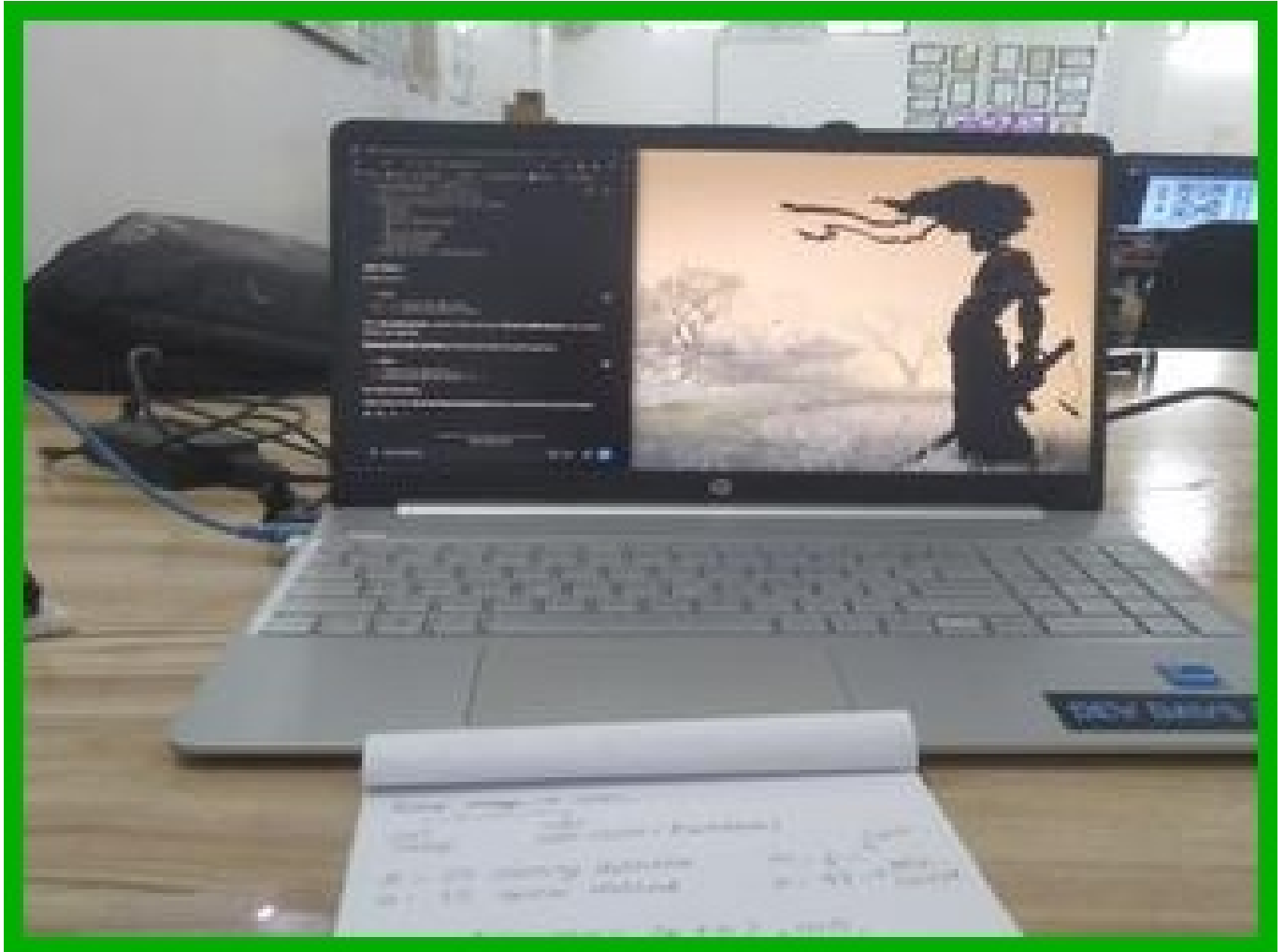


Image: noncattle_008.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.469 ms

Running Accuracy: 94.74%

CORRECT



Image: noncattle_009.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.582 ms

Running Accuracy: 94.87%

CORRECT



Image: noncattle_010.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.547 ms

Running Accuracy: 95.00%

CORRECT



Image: noncattle_011.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.522 ms

Running Accuracy: 95.12%

CORRECT



Image: noncattle_012.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.681 ms

Running Accuracy: 95.24%

CORRECT



Image: noncattle_013.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.801 ms

Running Accuracy: 95.35%

CORRECT



Image: noncattle_014.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.546 ms

Running Accuracy: 95.45%

CORRECT



Image: noncattle_015.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.697 ms

Running Accuracy: 95.56%

WRONG



Image: noncattle_016.jpg

Ground Truth: non_cattle

Prediction: cattle

Confidence: 55.08%

Cattle Boxes: 1

Latency: 1.619 ms

Running Accuracy: 93.48%

CORRECT



Image: noncattle_017.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 2.054 ms

Running Accuracy: 93.62%

CORRECT



Image: noncattle_018.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.751 ms

Running Accuracy: 93.75%

CORRECT



Image: noncattle_019.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 2.023 ms

Running Accuracy: 93.88%

CORRECT



Image: noncattle_020.jpg

Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 1.690 ms

Running Accuracy: 94.00%